Task 1: Provisioning an Amazon RDS Instance

Overview

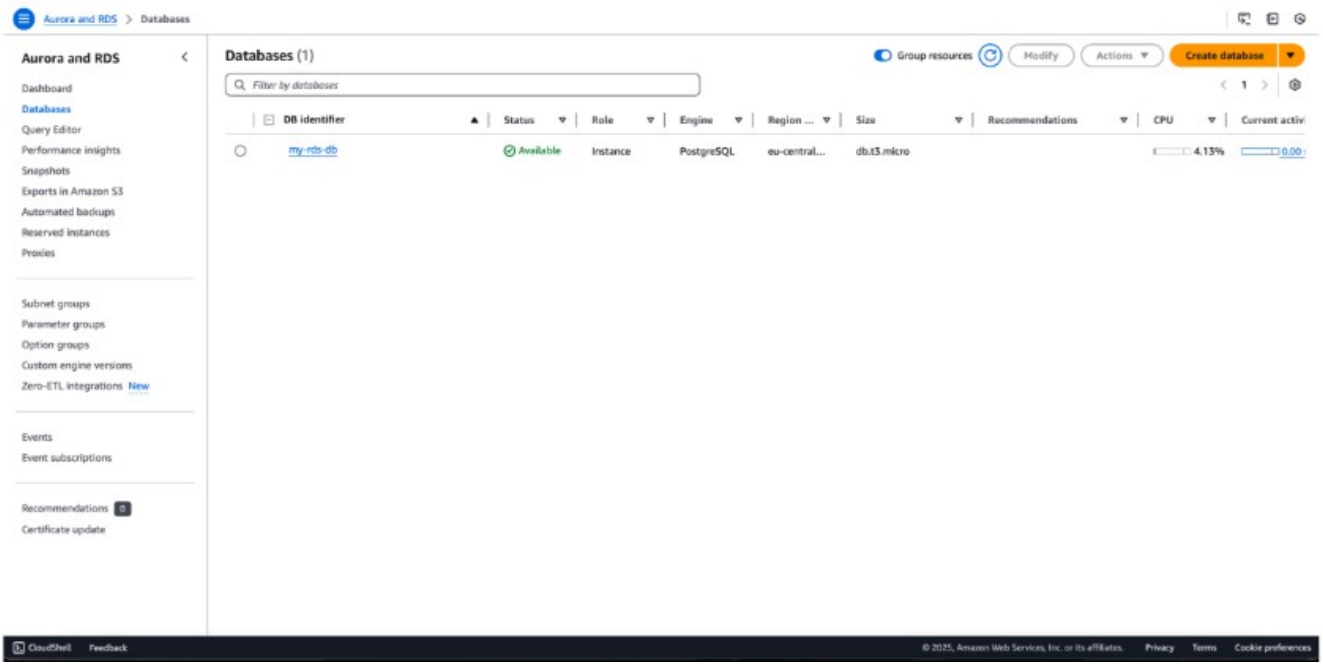
In this step, I deployed a managed relational database using **Amazon RDS (Relational Database Service)**. Using a managed service instead of manual installation on an EC2 instance reduces operational overhead and ensures better reliability.

Technical Specifications

- **Database Engine:** PostgreSQL (an open-source object-relational database).
- **Instance Class:** `db.t3.micro` (ideal for testing and low-traffic applications within the Free Tier).
- **Storage:** 20 GB General Purpose SSD (gp3).
- **Region:** eu-central-1 (Frankfurt).

Why this matters for Security & Operations:

- **Managed Updates:** AWS handles OS patching and database software updates, reducing the risk of vulnerabilities.
- **Isolation:** By placing the database in a specific subnet, we can control traffic using **Security Groups**, ensuring that only authorized applications (like a web server) can connect to it.
- **Scalability:** The `t3.micro` instance is a cost-effective starting point, but RDS allows for easy scaling as data requirements grow.



Task 2: Identifying the Database Endpoint

Overview

To connect any application to the database, we need its **Endpoint**. Think of the endpoint as the specific "home address" of your database on the internet (or within your private network).

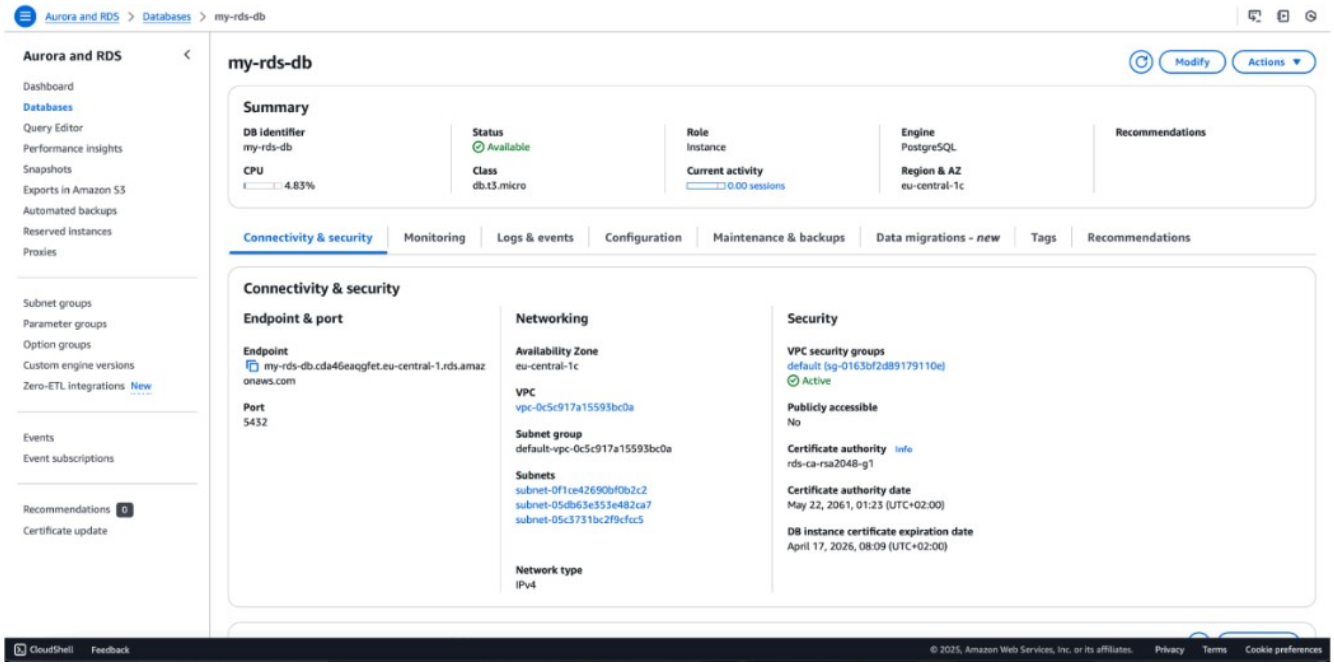
Key Details

- **Endpoint:** This is a DNS name (e.g., `my-rds-db.c123456789.eu-central-1.rds.amazonaws.com`).
- **Port:** Default for PostgreSQL is **5432**.

Security Perspective

Finding the endpoint is only the first step. In a secure environment:

1. **Private Access:** The endpoint should ideally not be reachable from the public internet.
2. **Firewall (Security Groups):** Even if someone has the endpoint address, they cannot connect unless their IP address is explicitly allowed in the **AWS Security Group** rules.



Task 3: Connecting via Command Line (psql)

Overview

Now that the database is running and the endpoint is known, I established a connection using **psql**, the command-line interface for PostgreSQL. This step verifies that the database is reachable and accepting connections.

The Command

I used the following command structure to log in: `psql -h <endpoint> -U <username> -d postgres`

Security & Educational Note

- **Encrypted Connection:** As seen in the screenshot, the connection uses **SSL (TLSv1.3)**. This is crucial for cybersecurity because it encrypts the data in transit, preventing attackers from "sniffing" your database password or data over the network.
- **Authentication:** Access is granted only after providing the correct password for the `databaseuser`.

- **Success Indicator:** The `postgres=>` prompt confirms that we are inside the database and ready to execute SQL queries.

```
> psql -h my-rds-db.cda46eaqgfet.eu-central-1.rds.amazonaws.com -U databaseuser -d postgres

Password for user databaseuser:
psql (14.17 (Homebrew), server 17.2)
WARNING: psql major version 14, server major version 17.
         Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

postgres=>
```

Task 4: Creating the Database Schema

Overview

Once connected, I defined the database structure by creating a table named **`users`**. This sets the foundation for how data will be stored and organized.

Table Structure Breakdown

- **`id`**: A unique identifier for each user, automatically incremented (**SERIAL PRIMARY KEY**).
- **`name`**: Stores the user's name (up to 100 characters).
- **`email`**: Stores the email address with a **UNIQUE** constraint.

Security & Best Practices

- **Data Integrity:** The `UNIQUE` constraint on the email field is a critical security and operational feature. It prevents duplicate accounts and ensures that each user has a distinct identity.
- **Least Privilege:** Defining specific data types (like `VARCHAR(100)`) instead of unlimited text helps optimize performance and prevents certain types of buffer-related issues.

```
postgres=> CREATE TABLE users (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);
CREATE TABLE
postgres=>
```

Task 5: Populating the Database (Data Entry)

Overview

To verify that the database structure is functional, I inserted a test record into the **`users`** table. This step confirms that the system correctly handles write operations and follows the defined constraints.

The Query

I executed the following command to add a new user: `INSERT INTO users (name, email) VALUES ('Іван', 'ivan@example.com');`

Technical Observations

- **Success Message:** The response **`INSERT 0 1`** indicates that one row was successfully added to the table.
- **Automatic ID Generation:** Since the `id` field was set to **SERIAL**, the database automatically assigned a unique number to this record behind the scenes.
- **UTF-8 Support:** The successful insertion of the name "Іван" confirms that the database correctly handles Cyrillic characters and Unicode encoding.

Cybersecurity Note

In a real-world application, data is never inserted directly like this from a user form. To prevent **SQL Injection** attacks, developers must use **Parameterized Queries** (Prepared Statements) to ensure that malicious code cannot be executed through input fields.

```
postgres=> INSERT INTO users (name, email) VALUES ('Іван', 'ivan@example.com');
INSERT 0 1
postgres=>
```

Task 6: Querying and Verifying Data

Overview

The final step is to retrieve the data from the database to ensure it was stored correctly. I used the **`SELECT`** statement, which is the primary tool for reading data in SQL.

The Query

I executed: `SELECT * FROM users;` *(The asterisk `*` tells the database to return all columns from the specified table).*

Technical Results

- **Confirmation:** The terminal successfully displayed the record for "Іван" with his email and the automatically assigned ID of **1**.
- **Data Persistence:** This confirms that the RDS instance is not only accepting connections but is actively storing and retrieving data as expected.

Cybersecurity Tip

From a security standpoint, it is best practice to **select only the specific columns you need** (e.g., `SELECT name FROM users;`) rather than using `SELECT *`. This reduces the amount of sensitive data being sent over the network and minimizes the risk of accidental information disclosure.

```
postgres=> SELECT * FROM users;
 id | name |      email
----+------+-----------------
  1 | Іван | ivan@example.com
(1 row)
```

Task 7: Resource Cleanup (Deleting the RDS Instance)

Overview

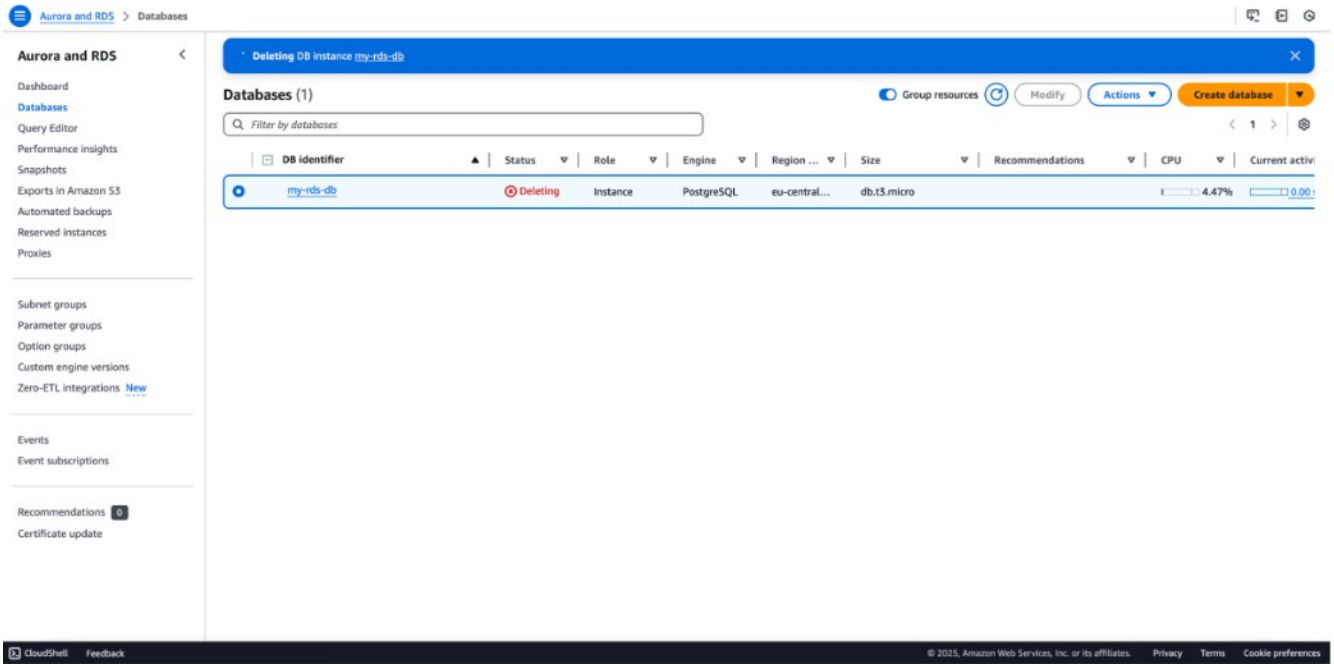
The final step of this laboratory work was the secure decommissioning of the cloud resources. Deleting unused instances is a fundamental practice in cloud management to avoid unnecessary costs and minimize security risks.

Key Actions

- **Termination:** I initiated the deletion process through the Amazon RDS console.
- **Final Snapshot (Optional):** During deletion, AWS offers the option to create a final backup (snapshot). For this lab, I proceeded without a snapshot to fully clear the environment.

Why this matters for Cybersecurity & Operations:

- **Cost Management:** In a professional environment, "zombie" resources (unused but active) can lead to massive unexpected bills.
- **Security Hygiene:** Every active resource is a potential target. Deleting an instance after a project is finished ensures that an unmonitored database doesn't become a vulnerability in the future.
- **Compliance:** Proper disposal of data and resources is a requirement for many security frameworks (like SOC2 or ISO 27001).


Aurora and RDS > Databases
Aurora and RDS
Dashboard
Databases
Query Editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations New
Events
Event subscriptions
Recommendations 0
Certificate update
Deleting DB instance my-rds-db
Databases (1)
Group resources
Modify
Actions
Create database
Filter by databases
DB identifier
Status
Role
Engine
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